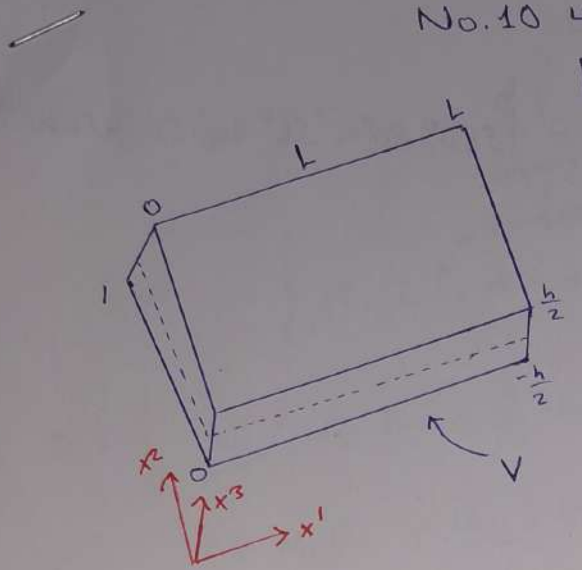
No.10 4.12.2024

$\boxed{1}$



Одномерная задача

$\bar{u} = \bar{u}(x^1, x^3)$

$$\delta W = \int_V \sigma_{ij}\, \delta\varepsilon_{ij}\, dV$$

$\cancel{\varepsilon_{12} = \varepsilon}$

$\varepsilon_{22}, \varepsilon_{12}, \varepsilon_{23} = 0$

$\sigma_{33} = 0$

$$\delta W = \int_V \sigma_{ij}\, \delta\varepsilon_{ij}\, dV$$

$\sigma_{13}\,\delta\varepsilon_{13} + \sigma_{31}\,\delta\varepsilon_{31} = \sigma_{13}\,\delta_{13} + \sigma_{13}\,\delta\varepsilon_{13} = 2\sigma_{13}\,\delta\varepsilon_{13}$

$$= \int_V \sigma_{11}\,\delta\varepsilon_{11} + 2\,\sigma_{13}\,\delta\varepsilon_{13}\, dV$$

$$\boxed{\begin{aligned} &\varepsilon_{11} = \overset{\circ}{u}_{1,1} + x^3 \overset{1}{u}_{1,1} \\ &\varepsilon_{13} = \frac{1}{2}(\overset{1}{u}_1 + \overset{\circ}{u}_{3,1}) \\ &\sigma_{11} = \frac{E}{1-\nu^2}(\overset{\circ}{u}_{1,1} + x^3 \overset{1}{u}_{1,1}) \\ &\sigma_{13} = \frac{E}{1+\nu}\cdot\frac{1}{2}(\overset{1}{u}_1 + \overset{\circ}{u}_{3,1}) \end{aligned}}$$

$$= \int_0^L \int_{-\frac{h}{2}}^{\frac{h}{2}} \int_0^1 \frac{E}{1-\nu^2}\,\delta(\overset{\circ}{u}_{1,1} + x^3 \overset{1}{u}_{1,1})^2 + \frac{E}{2(1+\nu)}\,\delta(\overset{1}{u}_1 + \overset{\circ}{u}_{3,1})^2\, dx^2\, dx^3\, dx^1$$

$$= \int_0^L \int_{-\frac{h}{2}}^{h/2} \frac{E}{1-\nu^2}\left(\overset{\circ}{u}_{1,1}\,\delta\overset{\circ}{u}_{1,1} + (x^3)^2\, \overset{1}{u}_{1,1}\,\delta\overset{1}{u}_{1,1} + \right.$$
$$\frac{E}{2(1+\nu)}(\overset{1}{u}_1 + \overset{\circ}{u}_{3,1})(\delta\overset{1}{u}_1 + \delta\overset{\circ}{u}_{3,1})\, dx^3\, dx^1$$

$$= \int_0^L \frac{Eh}{1-\nu^2}\left(\overset{\circ}{u}_{1,1}\,\delta\overset{\circ}{u}_{1,1} + \frac{h^2}{12}\,\overset{1}{u}_{1,1}\,\delta\overset{1}{u}_{1,1}\right) + \frac{Eh}{2(1+\nu)}(\overset{1}{u}_1 + \overset{\circ}{u}_{3,1})(\delta\overset{1}{u}_1 + \delta\overset{\circ}{u}_{3,1})\, dx^1$$

$$= \left[\frac{Eh}{1-\nu^2}\left(\overset{\circ}{u}_{1,1}\,\delta\overset{\circ}{u}_1 + \frac{h^2}{12}\,\overset{1}{u}_{1,1}\,\delta\overset{1}{u}_1\right) + \frac{Eh}{2(1+\nu)}(\overset{1}{u}_1 + \overset{\circ}{u}_{3,1})(\delta\overset{\circ}{u}_3)\right]\Bigg|_0^L$$

integration by parts

$$- \int_0^L \left[\frac{Eh}{1-\nu^2}\left(\overset{\circ}{u}_{1,11}\,\delta\overset{\circ}{u}_1 + \frac{h^2}{12}\,\overset{1}{u}_{1,11}\,\delta\overset{1}{u}_1\right) + \frac{Eh}{2(1+\nu)}(\overset{1}{u}_{1,1} + \overset{\circ}{u}_{3,11})\,\delta\overset{\circ}{u}_3\right.$$
$$\left. - \frac{Eh}{2(1+\nu)}(\overset{1}{u}_1 + \overset{\circ}{u}_{3,1})\,\delta\overset{1}{u}_1\right] dx^1$$

$$= \int_0^L \rho F_3\,\delta\overset{\circ}{u}_3\, dx^1 + \left(\overset{\circ}{M}_1\,\delta\overset{\circ}{u}_1 + \overset{1}{M}_1\,\delta\overset{1}{u}_1 + \overset{\circ}{M}_3\,\delta\overset{\circ}{u}_3\right)\Bigg|_0^L$$

$\boxed{2}$

$$M_i^j = \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho_i (x^3)(x^3)^i \, dx^3$$

$$\delta W_{ij} = \delta F_{масс} + \delta F_{повер.} = \int_V \rho F_i \delta u_i \, dV + \int_{\partial T} \rho_i \delta u_i \, dS$$

читается как внешняя сила

+ инерция

$$\delta W = \int_0^L \rho F_3 \delta u_3^0 \, dx^1 + \left(M_1^0 \delta u_1^0 + M_1^1 \delta u_1^1 + M_3^0 \delta u_3^0\right)\Big|_0^L$$

$$0 = \left[\frac{-Eh}{1-\nu^2}\left(u^0_{1,1} \delta u_1^0 + \frac{h^2}{12} u^1_{1,1} \delta u_1^1\right) + \frac{Eh}{2(1+\nu)}\left(u_1^1 + u^0_{3,1}\right) \delta u_3^0 + M_1^0 \delta u_1^0 + M_1^1 \delta u_1^1 + M_3^0 \delta u_3^0\right]\Big|_0^L + \int_0^L \Big[\frac{Eh}{1-\nu^2}\left(u^0_{1,11} \delta u_1^0 + \frac{h^2}{12} u^1_{1,11} \delta u_1^1\right) +$$

$$\frac{Eh}{2(1-\nu)}\left(u^1_{1,1} + u^0_{3,1}\right) \delta u_3^0 - \frac{Eh}{2(1+\nu)}\left(u_1^1 + u^0_{3,1}\right) \delta u_1^1 + \underline{\rho F_3 \, \delta u_3 \delta u_3^0} \ ?$$

$$- \rho h \ddot u_1^0 \delta u_1^0 - \frac{\rho h^3}{12} \ddot u_1^1 \delta u_1^1 - \rho h \ddot u_3^0 \delta u_3^0 \Big] \, dx^3$$

$$\left[\left(M_1^0 - \frac{Eh}{1-\nu^2} u^0_{1,1}\right) \delta u_1^0 + \left(M_1^1 - \frac{Eh^3}{12} u^1_{1,1}\right) \delta u_1^1 + M_3^0 - \frac{Eh}{2(1+\nu)}\left(u_1^1 + u^0_{3,1}\right) \delta u_3^0 \right|_0^L$$

$$+ \int_0^L \Big\{ \left(\frac{Eh}{1-\nu^2} u^0_{1,11} - \rho h \ddot u_1^0\right) \delta u_1^0 + \left(\frac{Eh}{2(1+\nu)}\left(u^1_{1,1} + u^0_{3,11}\right) + \rho F_3 - \rho h \ddot u_3^0\right) \delta u_3^0$$

$$+ \left(\frac{Eh^3}{12(1-\nu^2)} u^1_{1,11} - \frac{Eh}{2(1+\nu)}\left(u_1^1 + u^0_{3,1}\right) - \rho \frac{h^3}{12} \ddot u_1^1\right) \delta u_1^1 \Big\} \, dx^1 = 0$$

Гран. условие:

$$\frac{Eh}{1-\nu^2} u^0_{1,1} = M_1^0, \quad \frac{Eh^3}{12} u^1_{1,1} = M_1^1$$

$$\frac{Eh}{2(1+\nu)}\left(u_1^1 + u^0_{3,1}\right) = M_3^0$$

Система уравнений:

можно сократить все ур-ие

$$\begin{cases} \frac{Eh}{1-\nu^2} u^0_{1,11} = \rho h \ddot u_1^0 \\ \frac{Eh}{2(1+\nu)}\left(u^0_{3,11} + u^1_{1,1}\right) + \rho F_3 = \rho h \ddot u_3^0 \quad 3 \ ? \\ \frac{Eh^3}{12(1-\nu^2)} u^1_{1,11} - \frac{Eh}{2(1+\nu)}\left(u^0_{3,1} + u_1^1\right) = \rho \frac{h^3}{12} \ddot u_1^1 \end{cases}$$

$$\begin{cases} \frac{E}{1-\nu^2} \overset{0}{u}_{1,11} = \rho \ddot{\overset{0}{u}}_1 \\ \frac{E}{2(1+\nu)} (\overset{0}{u}_{3,11} + \overset{1}{u}_{1,1}) + \rho F_3 = \rho h \ddot{\overset{0}{u}}_3 \\ \frac{E}{1-\nu^2} \overset{1}{u}_{1,11} - \frac{E \cdot 6}{h^2(1+\nu)} (\overset{0}{u}_{3,1} + \overset{1}{u}_1) = \rho \ddot{\overset{1}{u}}_1 \end{cases}$$

← Волновое ур. $c^2 = \frac{E}{\rho(1-\nu^2)}$ $u_{tt} = c^2 u_{xx}$

$$\begin{cases} \frac{E}{1-\nu^2} \overset{1}{u}_{1,11} - \frac{6E}{h^2(1+\nu)} (\overset{0}{u}_{3,1} + \overset{1}{u}_1) = \rho \ddot{\overset{1}{u}}_1 \\ k\frac{E}{2(1+\nu)} (\overset{0}{u}_{3,11} + \overset{1}{u}_{1,1}) + \rho F_3 = \rho \ddot{\overset{0}{u}}_3 \end{cases}$$

Безразмерные величины

L - длина

$x = x^1/L$, $w = \overset{0}{u}_3/L$, $\psi = \overset{1}{u}_1$

$\eta = h/L$, $c = \sqrt{\frac{E}{\rho(1-\nu^2)}}$ $t^* = \frac{tc}{L}$

$\sigma_{33} = \overset{0}{\sigma}_{33} \cdot f_3(x_3)$ $\int_{-h/2}^{h/2} f(x^3)\, dx^3 = k^2$

t

4

$$\begin{cases} a\left(\dfrac{\partial^2 w}{\partial x^2} + \dfrac{\partial \psi}{\partial x}\right) + \rho F_3 = \dfrac{\partial^2 w}{\partial t^2} \\ \dfrac{\partial^2 \psi}{\partial x^2} - \dfrac{12}{\eta^2} a\left(\dfrac{\partial w}{\partial x} + \psi\right) = \dfrac{\partial^2 w}{\partial t^2} \end{cases}$$

2nd equ

diff $\dfrac{\partial}{\partial x}$ → 2nd eq

$$a = \frac{k^2(1-\nu)}{2}$$

$$\begin{cases} a\left(\dfrac{\partial^2 w}{\partial x^2} + \dfrac{\partial \psi}{\partial x}\right) = \dfrac{\partial^2 w}{\partial t^2} \\ \dfrac{\partial^2 \psi}{\partial x^2} - \dfrac{12a}{\eta^2}\left(\dfrac{\partial w}{\partial x} + \psi\right) = \dfrac{\partial^2 \psi}{\partial t^2} \end{cases}$$

$$u = c\, e^{i(\alpha x + wt)}$$

$$w = A \sin(\alpha x + wt) \qquad \psi = B \cos(\alpha x + wt)$$

$$a\left[\left(-A\alpha^2 \sin(\alpha x + wt) - B\alpha \sin(\alpha x + wt)\right)\right] = -A w^2 \sin(\alpha x + wt)$$

$$-B\alpha^2 \cos(\alpha x + wt) - \frac{12a}{\eta^2}\left(A\alpha \cos(\alpha x + wt) + B\cos(\alpha x + wt)\right) = -B w^2 \cos(\alpha x + wt)$$

$$\begin{cases} a(-A\alpha^2 - B\alpha) = -A w^2 \\ -B\alpha^2 - \dfrac{12}{\eta^2} a(A\alpha + B) = -B w^2 \end{cases}$$

$$K = \frac{A}{B}$$

$$K = \frac{A}{B}$$

$$\begin{cases} a(K\alpha^2 + \alpha) = K w^2 \\ \alpha^2 + \dfrac{12a}{\eta^2}(K\alpha + 1) = w^2 \end{cases}$$

$$k\alpha^2 + \frac{12a}{\eta^2}\left(k^2\alpha + k\right) = a\left(k\alpha^2 + \alpha\right)$$

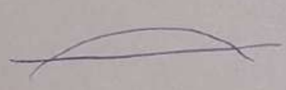
квадратное ур-е относительно k

$$\Rightarrow \frac{12a\alpha}{\eta^2}k^2 + \left[\alpha^2(1-a) + \frac{12a}{\eta^2}\right]k - a\alpha = 0$$

$$\Rightarrow k^2 + \left[\frac{\eta^2(1-a)\alpha}{12a} + \frac{1}{\alpha}\right]k - \cancel{\alpha}\frac{\eta^2}{12} = 0 \qquad \alpha \to 0$$

$$\left.\begin{array}{l} k_1 + k_2 \to \infty \\ k_1 \cdot k_2 = \cancel{\alpha^2}\dfrac{\eta^2}{12} \end{array}\right\} \quad \begin{array}{ll} k_1 \to \infty & \dfrac{A}{B} \to \infty \\ k_2 \to 0 & \dfrac{A}{B} \to 0 \end{array}$$

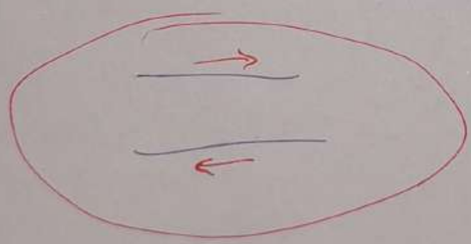
$\frac{A}{B} \to \infty$



прогибы большие

поворот маленький

$\frac{A}{B} \to$



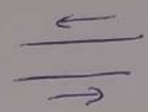
ИММ-С

$$\omega^4 \mp P\omega^2 + q = 0$$ ур-е отнсительно ω

$$\omega_{1,2}^2 = \frac{P}{2} \pm \sqrt{\frac{P^2}{4} - q}$$

решение квадр. ур.

$$\omega_1^2 = \frac{P}{2} + \sqrt{\frac{P^2}{4}}$$



$$\omega_2^2 = \frac{P^2}{2} \mp \sqrt{\frac{P^2}{4}}$$

МКР:

$$\boxed{\begin{aligned} &a(D_{11} w + D_{01}\psi) = D_{tt} w \\ &D_{11}\psi - \frac{12a}{\eta^2}(D_{01} w + \psi) = D_{tt}\psi \end{aligned}}$$

В-РМ:

$$\boxed{\begin{aligned} &a D_{11} w + D_{01}\psi = D_{tt} w \\ &D_{11}\psi - \frac{12a}{\eta^2}(D_{01} w + D_{00}\psi) = D_{tt}\psi \end{aligned}}$$

МКЭ:

$$\boxed{\begin{aligned} &a(D_{11} w + D_{01}\psi) = D_{tt} w \\ &D_{11}\psi - \frac{12a}{\eta^2}(D_{01} w + D_{00}^{*}\psi) = D_{tt}\psi \end{aligned}}$$

$$D_{11} f_j = \frac{1}{h^2}(f_{j+1} - 2f_j + f_{j-1})$$

$$D_{01} f_j = \frac{1}{2h}(f_{j+1} - f_{j-1})$$

$$D_{00} f_j = \frac{1}{4}(f_{j-1} + 2f_j + f_{j+1})$$

$$D_{00}^{*} f_j = \frac{1}{6}(f_{j-1} + 4f_j + f_{j-1})$$

$$D_{tt} f^k = \frac{1}{\tau^2}(f^{k+1} - 2f^k + f^{k-1})$$

MKP $\quad D_a \Psi = \frac{1}{a} D_{tt} w - D_{11} w$

$$\frac{\partial \Psi}{\partial x} = \frac{1}{a}\frac{\partial^2 w}{\partial t^2} - \frac{\partial^2 w}{\partial x^2}$$

$$\frac{12a}{\eta^2}$$

Differentiation the second equation by $\frac{\partial}{\partial x}$:

$$\frac{1}{a}\frac{\partial^4 w}{\partial x^2 \partial t^2} - \frac{\partial^4 w}{\partial x^4} - \left(\cancel{\frac{\partial^2 w}{\partial x^2}} + \frac{1}{a}\frac{\partial^2 w}{\partial t^2} - \cancel{\frac{\partial^2 w}{\partial x^2}} \right)$$

in page 4

$$= \frac{1}{a}\frac{\partial^4 w}{\partial t^4} - \frac{\partial^4 w}{\partial x^2 \partial t^2}$$

$$\frac{\partial^4 w}{\partial x^4} - \left(1 + \frac{1}{a}\right)\frac{\partial^4 w}{\partial x^2 \partial t^2} + \frac{12}{\eta^2}\frac{\partial^2 w}{\partial t^2} + \frac{1}{a}\frac{\partial^4 w}{\partial t^4} = 0$$